Name:

BUID:

Workshop 3: The λ-Calculus and Evaluation Strategies

1. Write `fun x -> x x y (y z w) x (fun x -> x y z)` in the compact abstract syntax of the λ-calculus.
2. Rewrite $\lambda x.abc\lambda y.xyz$ with all necessary parentheses (i.e., assuming no precedence or associativity).
3. Determine a value v such that $(\lambda x.xx)(\lambda y.y) \longrightarrow^\star v$ and provide a derivation using the full non-deterministic semantics.
4. Determine $\mathsf{FV}(e)$ and $\mathsf{BV}(e)$ where e is $\lambda x.\lambda v.(\lambda y.yx)(\lambda y.wy)(\lambda w.w)$. Additionally, draw an arrow from each bound variable to the binder that binds it.
5. Perform the substitution $[\lambda y.y/x]((\lambda x.xy)(\lambda y.x))$ with capture avoidance.
6. Perform the substitution $[y/x](\lambda y.xy)$ with capture avoidance.
7. Determine an expression e such that $(\lambda x.x)((\lambda z.z)(\lambda w.w)) \longrightarrow e$ under CBV and CBN, and provide a derivation.
8. Derive $(\lambda x.\lambda y.y)((\lambda z.z)(\lambda q.q)) \Downarrow \lambda y.y$ under CBV and CBN.
9. Determine an closed expression that terminates under CBN but not under CBV.
10. **(Challenge)** Determine closed expressions $\{e_i\}_{i=1}^{\infty}$ such that $|e_n| = O(n)$, CBN requires $\Omega(2^n)$ steps, and CBV takes $O(n)$ steps.
11. **(Challenge)** Determine closed expressions $\{e_i\}_{i=1}^{\infty}$ such that $|e_n| = O(n)$ and $e_n \longrightarrow^\star e$ under CBV, where $|e| = \Omega(2^n)$.
12. **(Challenge)** Do the previous problem again, but further require that the normal form of e_n is $\lambda x.x$ and that it takes $O(n)$ steps to evaluate.[1]

Write your solutions below and on the back of this page.

[1] It's worth thinking about why this matters; feel free to ask about it during the workshop.

Continue your work here.

Non-deterministic full small-step semantics for the λ-calculus.

$$\frac{e_1 \longrightarrow e_2}{e_1 e \longrightarrow e_2 e}\ \textsc{appLeft} \qquad \frac{e_1 \longrightarrow e_2}{e e_1 \longrightarrow e e_2}\ \textsc{appRight} \qquad \frac{e_1 \longrightarrow e_2}{\lambda x.e_1 \longrightarrow \lambda x.e_2}\ \textsc{fun} \qquad \frac{}{(\lambda x.e_1)e_2 \longrightarrow [e_2/x]e_1}\ \textsc{beta}$$

Capture-avoiding substitution.

$$[e/x](\lambda y.e_1) = \begin{cases} \lambda y.e_1 & x = y \\ \lambda y.[e/x]e_1 & x \neq y, y \notin \mathsf{FV}(e) \\ \lambda z.[e/x][z/y]e_1 & x \neq y, y \in \mathsf{FV}(e), z \text{ is fresh} \end{cases}$$

$$[e/x](e_1 e_2) = ([e/x]e_1)([e/x]e_2)$$

$$[e/x]y = \begin{cases} e & x = y \\ y & x \neq y \end{cases}$$

Call-by-value small-step semantics for the λ-calculus.

$$\frac{e_1 \longrightarrow e_2}{e_1 e \longrightarrow e_2 e}\ \textsc{appLeft} \qquad \frac{e_1 \longrightarrow e_2}{(\lambda x.e)e_1 \longrightarrow (\lambda x.e)e_2}\ \textsc{appRight} \qquad \frac{}{(\lambda x.e_1)(\lambda y.e_2) \longrightarrow [\lambda y.e_2/x]e_1}\ \textsc{beta}$$

Call-by-name small-step semantics for the λ-calculus.

$$\frac{e_1 \longrightarrow e_2}{e_1 e \longrightarrow e_2 e}\ \textsc{appLeft} \qquad \frac{}{(\lambda x.e_1)e_2 \longrightarrow [e_2/x]e_1}\ \textsc{beta}$$

Call-by-value big-step semantics for the λ-calculus.

$$\frac{}{\lambda x.e \Downarrow \lambda x.e}\ \textsc{fun} \qquad \frac{e_1 \Downarrow \lambda x.e \qquad e_2 \Downarrow v_2 \qquad [v_2/x]e \Downarrow v}{e_1 e_2 \Downarrow v}\ \textsc{app}$$

Call-by-name big-step semantics for the λ-calculus.

$$\frac{}{\lambda x.e \Downarrow \lambda x.e}\ \textsc{fun} \qquad \frac{e_1 \Downarrow \lambda x.e \qquad [e_2/x]e \Downarrow v}{e_1 e_2 \Downarrow v}\ \textsc{app}$$